

Better Questions.

Better Matches.

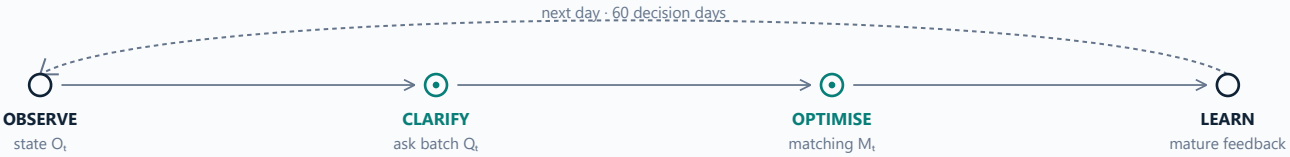
Allocation-Aware Clarification and Global Reciprocal Matching Under Uncertainty

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RESEARCH HYPOTHESIS

Proposed, not established. Assessed by paired simulator evaluation and exact validity checks [1].

Problem. Clarification decisions and introduction allocation are interdependent under uncertainty and reciprocal hard constraints.
Mechanism. We propose a feasibility-first policy that prioritises questions by their estimated effect on the best feasible matching, values feasible edges by downstream MSMI, and selects introductions jointly through maximum-weight general-graph matching.
Testable claim. We hypothesise that the integrated policy improves family-averaged MSMI per 100 arrived members over the official greedy, no-clarification and random-feasible baselines. Paired simulator evaluation may falsify this claim; improvement is not assumed.



01 GATE

Hard constraints define what may be done. A score never overrides them.

02 ASK

Information is valuable only if it can change today's best allocation.

03 ALLOCATE

Introductions compete for people; choose them together, not one by one.

All profiles and outcomes in the challenge environment are synthetic. This note proposes a method and a pre-specified evaluation plan; it reports no experimental results. Rules follow the official participant specification, release 1.0.0 (5 October 2026) [1].

PROBLEM FORMULATION

The Sequential Decision Problem

The challenge is a finite-horizon policy problem, not independent pair ranking. On decision day t in {0,...,59}, the observable state is

$$O_t = (A_t, X_t, Z_t, H_t, B_t)$$

A_t contains arrived and available members; X_t observable profiles; Z_t field status and observation time; H_t released interaction history; and B_t remaining ask budget plus permitted memory. Future arrivals, hidden state, undisclosed answers and unreleased outcomes are excluded [1].

OFFICIAL HORIZON
60 decision days
40 follow-up days
12 ask units/day
Hard bundle: 3 units
Soft field: 1 unit
Response deadline: 7 days

Each day has two legal phases: submit one batch of clarification requests, receive results and refresh the observation, then submit a feasible non-overlapping introduction batch. There is no intra-day adaptive question-by-question loop.

Objective and Ranking

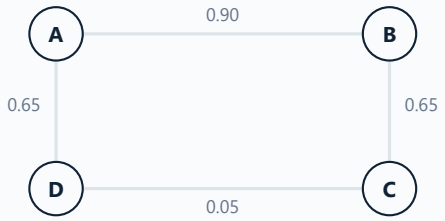
For episode s, let M_s be qualifying introductions and N_s members arrived by day 59:

$$R_s = 100 \times M_s / N_s$$

One qualifying pair counts once. The primary score averages episode scores within each of six families, then averages the six family means equally. Tie-breaks are coverage, mutual acceptances per 100, lower ask cost and lower inference time [1].

ENDPOINT COMPETITION - ILLUSTRATIVE ONLY

Pair	Value
A-B	0.90
C-D	0.05
A-D	0.65
B-C	0.65



Greedy total: 0.95 **Global total: 1.30**

Validity is lexicographic. A favourable score never overrides a hard constraint. An invalid ask, malformed response, runtime failure or invalid batch makes the assessed episode invalid and ineligible for ranking [1].

RESEARCH FOUNDATION

Established Tools, Challenge-Specific Integration

The proposal combines established ideas without claiming that their combination is already proven. Reciprocal recommendation motivates modelling value to both endpoints [2,3]; active feature acquisition motivates paying for information only when it can improve a downstream decision [5]; Edmonds supplies the general-graph matching abstraction [4]; selective-label and delayed-learning work motivate time-safe, cautious updating [6,8].

FOUNDATION	ESTABLISHED PRINCIPLE	OUR CHALLENGE-SPECIFIC USE
Reciprocal recommendation	People are both recommenders and recommendands; bilateral utility matters.	Symmetric pair/context features and a pair-level downstream outcome, never one-sided relevance.
Active feature acquisition	Information has value relative to a downstream decision and cost.	Approximate the change in the best feasible matching value under plausible clarification answers.
General-graph matching	A matching contains edges sharing no endpoint.	Jointly select introductions because preferences do not guarantee a fixed bipartition.
Sequential learning	Actions select labels; rewards can be delayed.	Train only on mature, released feedback; exploration remains bounded and optional.

One Organising Decision Principle



STATUS	MECHANISMS	BURDEN OF PROOF
CORE	Exact feasibility; time-safe features; conditional MSMI; maturity masks; calibration when useful.	Validity and basic fidelity.
PRIMARY	Allocation-aware clarification and maximum-weight general-graph matching.	Central hypothesis.
OPTIONAL	Bootstrap exploration; conservative recency adaptation; strategic waiting.	Include only if ablations help.

Scientific boundary. Prior work supports the components, not the success of this exact policy in this simulator. We propose a falsifiable integration and will evaluate it against the official baselines; we do not claim measured improvement.

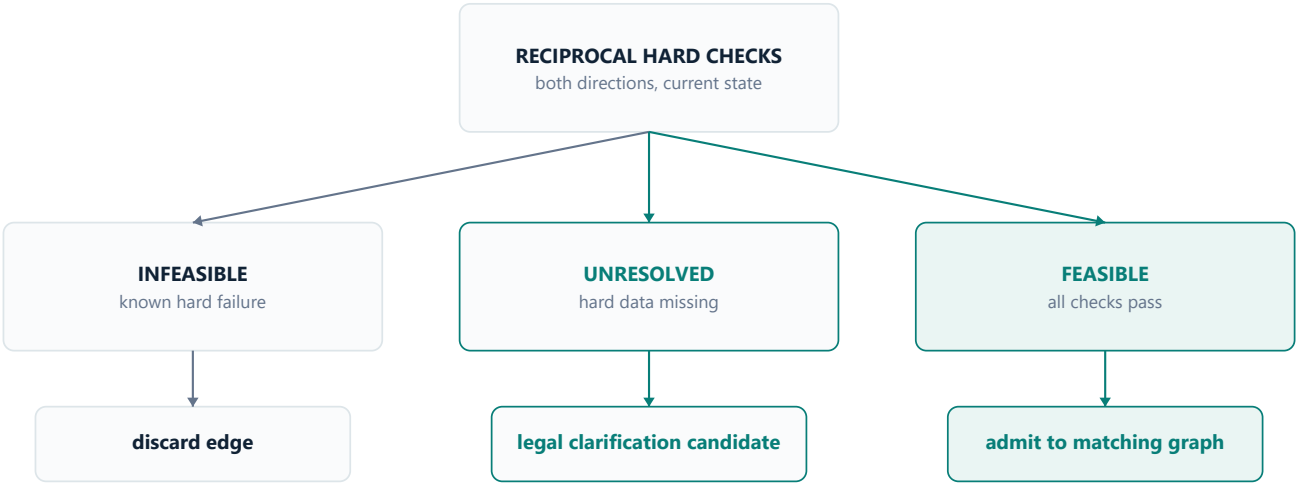
VALIDITY LAYER

Exact Reciprocal Feasibility

For any arrived candidate pair (i,j), the official eligibility semantics induce the three-state gate

$$g_t(i,j) \in \{\text{infeasible}, \text{unresolved}, \text{feasible}\}$$

A known hard failure dominates missingness. Unknown or declined mandatory information cannot be imputed into feasibility. Only currently feasible edges may enter the final matching graph [1].



Pair-Level Hard Checks

- Distinct adults in the same pool.
- Each age lies inside the other's stated range.
- Each stated gender is accepted by the other.
- Relationship structures agree.
- Reciprocal smoking and current-children constraints.
- No explicit yes/no conflict on wanting children.
- Each accepts the other's current zone.
- At least one overlapping schedule slot.

Operational Checks at Output

- Both members currently available.
- No outstanding concurrent introduction.
- Pair not previously introduced in the episode.
- Each endpoint used at most once in the batch.
- Batch shape and identifiers are valid.
- Empty graphs safely return an empty batch.

Atomicity matters. The simulator validates the complete batch before applying it; an invalid action creates no partial batch and invalidates the episode.

Time-safety rule. Reconstruct eligibility from values, field status and observation times available at day t. Never backfill an early decision with a later answer, private truth, generator seed or opaque-ID inference.

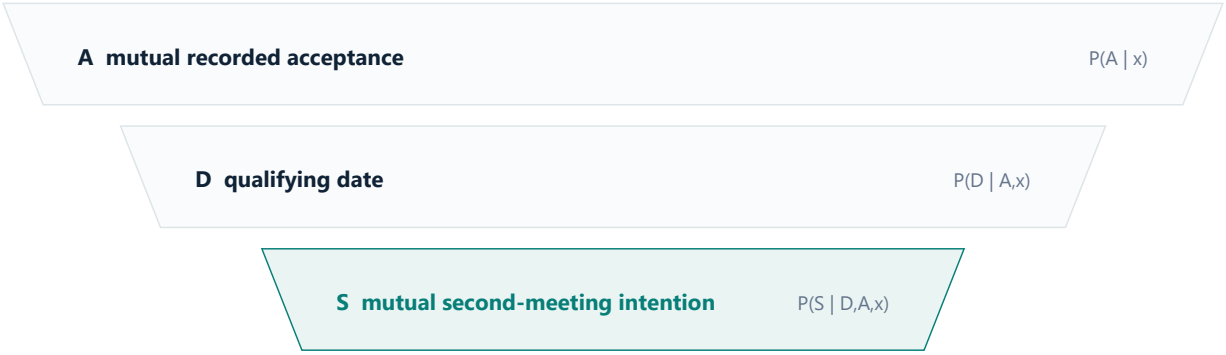
OUTCOME MODEL

Estimating Downstream MSMI

For each currently feasible edge e, estimate the probability of the exact downstream outcome through nested conditional events:

$$\hat{\mu}_e = \hat{P}(A \mid x_e) \times \hat{P}(D \mid A, x_e) \times \hat{P}(S \mid D, A, x_e)$$

A is recorded mutual acceptance; D is a qualifying date within 30 days of assignment; S is two timely 'Yes' second-meeting intentions. This is a chain-rule factorisation of nested events - not an independence claim about the two people [1].



Lightweight Model Stack

- Regularised logistic models as default; a small tree model as challenger.
- Reciprocal similarities/differences, age gap, schedule overlap and context.
- Missingness and disclosure-status indicators.
- Time since arrival, candidate scarcity and current graph degree.
- Shrunk member-history summaries using only released outcomes.

Training and Diagnostics

- Store action-time feature snapshots; never attach final profiles retrospectively.
- Train each stage only after its target window is mature.
- Fit population priors and partial-pooling shrinkage for sparse histories.
- Assess reliability, Brier score and log loss on held-out mature observations.
- Apply low-parameter calibration only if held-out probability quality improves [10].

Comparator, not assumption. A direct regularised MSMI classifier will be evaluated against the funnel. The funnel may use intermediate supervision; the direct model may avoid propagated stage error. Neither is declared superior before experiments.

Target-Specific Maturity Gates

A ACCEPTANCE	D DATE	S SECOND INTENTION
response window closed	qualifying-date window closed	date plus response window closed

PRIMARY CONTRIBUTION

Allocation-Aware Clarification

Let $V^*(O_t)$ be the estimated value of the best current feasible matching. For a legal question q with answer model $\hat{P}(a \mid O_t, q)$:

$$aVOI(q) = E_a[V^*(O_t \text{ after } q = a)] - V^*(O_t)$$
$$priority(q) = \max\{0, aVOI(q)\} / c(q)$$

Uncertainty alone is insufficient. A question matters only if plausible answers can unlock feasible edges, alter edge values, or change the coordinated allocation. Declined or undisclosed answers are explicit scenarios in the answer model. This is an approximation to decision-relevant information value, not an exact POMDP solution [5].



Cheap Screen

Hard-bundle asks are favoured when a member has several unresolved, otherwise promising edges, especially to scarce endpoints. Soft-field asks are considered near the allocation frontier: endpoints in the current matching or close high-value alternatives. Known hard failures are removed first.

Capped Hypothetical Evaluation

For a small shortlist, enumerate legal categorical answers or sample a few bundle outcomes from legitimately observed training rollouts. Cache unaffected work, cap total matching re-solves, then select a cost-constrained batch with a redundancy penalty.

Batched-interface compliance. All asks are chosen before any same-day answer is revealed. The policy refreshes once after the batch, then reconstructs exact feasibility before matching. No intra-day adaptive interrogation is assumed.

ALLOCATION LAYER

Global Reciprocal Matching

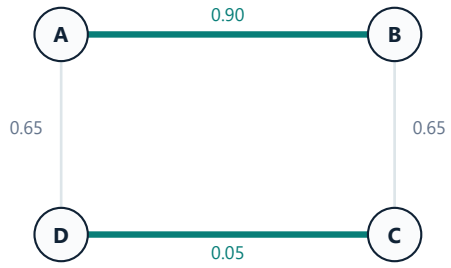
After clarification, construct the feasible general graph $G_t=(V_t,E_t)$ and assign each edge $\{i,j\}$ the calibrated downstream estimate $w_{ij}=\hat{\mu}_{ij}$. Select:

$$M_t^* = \arg\max_M \sum_{\{i,j\} \text{ in } M} w_{ij}$$

M is a matching on E_t; $\deg_M(i) \le 1$ for every i

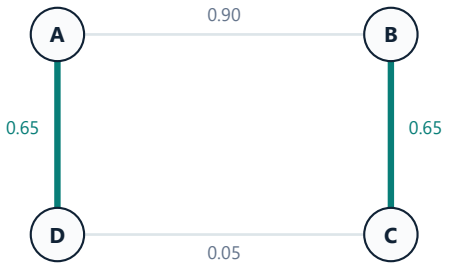
All availability, eligibility, repeat-pair and outstanding-introduction rules are enforced before output. Reciprocal gender preferences do not guarantee a fixed two-sided partition, so the graph must not be treated as bipartite by default. Edmonds-style general-graph matching is the correct abstraction [4].

Descending-edge greedy



Total = 0.95

Maximum-weight matching



Total = 1.30

Selected edges are teal; unselected alternatives are grey. Values are illustrative, not predictions or experimental results.

Why Greedy Can Lose

The highest single edge may consume two endpoints whose separate alternatives jointly create more value. Sorting edges is useful as a fallback, but it does not optimise the sum over a non-overlapping set.

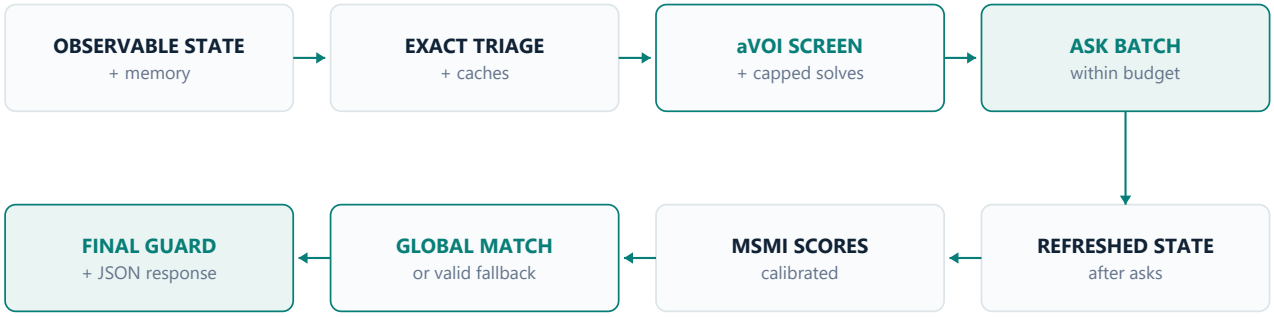
Operational Safeguards

Prune impossible pairs, score only feasible edges, solve components where useful, and enforce an internal solver deadline. If the bounded solve has not completed, use a precomputed valid greedy matching, then run the final guard.

Guarantee versus hypothesis. When the exact solver completes successfully, its result maximises total estimated edge weight for the graph it receives. A deadline-triggered greedy fallback remains feasible but is not guaranteed maximum-weight. Neither case guarantees actual downstream MSMI, that matching now beats waiting, or that the full sequential policy is optimal.

SYSTEM DESIGN

Complete Policy and Computational Implementation



Policy Skeleton

```
ON ASK(state, memory)
memory = update_observed(state,memory)
F,U = exact_triage(state)
if F and U are empty: return {"asks": [], "memory": memory}
w0 = cached_scores(F)
fallback0 = valid_greedy(F,w0)
base = bounded_matching(F,w0,ask_deadline,fallback0)
Q = cheap_screen(legal_asks,F,U,base)
Q = capped_hypothetical_value(topK(Q))
return {"asks": budgeted_batch(Q,12), "memory": memory}

ON MATCH(refreshed_state, memory)
memory = update_observed(refreshed_state,memory)
F,_ = exact_triage(refreshed_state)
if F is empty: return {"pairs": [], "memory": memory}
w = calibrated_funnel_scores(F)
fallback = valid_greedy(F,w)
M = bounded_matching(F,w,match_deadline,fallback)
return {"pairs": final_exact_guard(M), "memory": memory}
```

Runtime Discipline

- Hard-rule and coarse geography pruning before feature work.
- Cache member transforms, pair features and unchanged edge scores.
- Use low-dimensional models with bounded inference cost.
- Limit shortlist size, answer scenarios and matching re-solves.
- Give ASK and MATCH separate cancellable internal deadlines.
- Reserve headroom in both phases for validation, memory output and serialization.

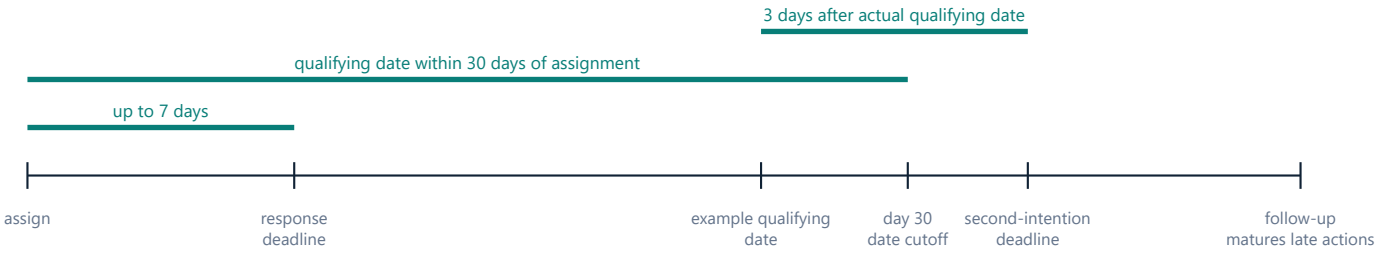
Official envelope. Each offline invocation includes process startup and must fit 10 seconds, 2 CPU cores, 1 GiB RAM, 64 processes and 1 MiB request/response/memory limits. Network and GPU access are unavailable. No latency is claimed here; worst-case container profiling is required [1].

Compute the deterministic feasible-greedy fallback before launching the bounded optimiser. If the internal solver deadline is reached, return that already available fallback, preserving time for the final guard and JSON serialization.

LEARNING LAYER

Delayed Feedback and Robust Updating

Assignments, directional responses, dates and second-meeting intentions are distinct events. The policy must learn from what has been released, not from what will eventually happen.



The day-30 cutoff is assignment-relative; the intention deadline is date-relative and may fall later.
Illustrative event timing; horizontal distances are not to scale.

Maturity, Censoring, Missingness

- A pending response is not a negative label.
- No response by a target deadline means the recorded event did not occur; it does not prove dislike.
- A pair that never dates has no observed second-meeting preference.
- Train a stage only when its window has closed and required history is visible.
- Retain late feedback for diagnostics without counting it toward on-time MSMI.

Selective Feedback and Drift

Only introduced pairs reveal downstream outcomes, so the policy shapes its own training data [6]. Historical propensities are null; unbiased inverse-propensity evaluation is not claimed. Diverse rollout policies and small feasibility-safe exploration may reduce lock-in [7]. Conservative recency weighting or a simple change detector can be tested for changing response conditions [11].

Time-safe update contract. Save the observation and features used at action time. Join only feedback whose observed_day is no later than the update day. Apply target-specific maturity masks. Reinitialise memory for each episode. Never carry evidence from private evaluation runs.

Optional Modules Must Earn Inclusion

EXTENSION	POTENTIAL VALUE	GUARDRAIL
Bootstrap exploration	Diversifies uncertain feasible edges and may reduce selective-label lock-in.	Small/decaying randomisation; strict feasibility; no-exploration ablation.
Recency adaptation	Responds to changing conditions with released intermediate feedback.	Conservative weighting; stable base; adaptation-off ablation.
Strategic waiting	Preserves option value for future arrivals or scarce members.	Default no-wait unless paired validation improves MSMI without unacceptable coverage loss.

Waiting is not assumed beneficial: dynamic matching theory makes its value conditional on market thickness, information and waiting costs [9].

EVIDENCE PLAN

Rigorous Evaluation Protocol

The central hypothesis will be tested by complete-policy simulator rollouts, never by fabricated results or row-wise splits of participant histories.

Data and Scenario Separation

- Keep the official six training, two validation and two development-test pools disjoint.
- Train, tune and freeze in that order; never split rows from the same people or pools across stages.
- Evaluate six families: standard (development in public code), sparse geography, cold start, delayed dates, shifted outcome weights and changing response conditions (drift).
- Use identical worlds and paired seeds for every method.
- Private design: 20 seeds/family, 120 episodes; local work uses multiple declared public seeds [1].

Official Comparisons

- Greedy: baseline asks plus descending feasible-edge selection.
- No clarification: the official no-asks baseline.
- Random feasible: shuffled feasible edges with non-overlap.

Our full policy and every ablation receive the same seeds, scenarios, horizon and follow-up. Each episode begins with empty memory. All methods must pass interface and validity checks before scores are compared.

Pre-Specified Measurement Stack

LAYER	MEASURE	INTERPRETATION
Primary	MSMI per 100 arrived; equal-weight mean across family means.	Direct ranking objective.
Service	Coverage; first-introduction wait; unserved count.	Detects gains from serving fewer people.
Funnel	Assignments, mutual acceptances, dates, MSMI, missing feedback.	Locates where outcomes change.
Information	Ask units, hard/soft mix, disclosure/decline outcomes.	Tests whether questions are decision-relevant.
Reliability	Calibration, Brier score, log loss on mature held-out labels.	Checks cardinal edge values.
Safety/compute	Invalid episodes, time, sizes, fallback rate.	Confirms deployability and eligibility.

Paired Uncertainty Analysis

Compute within-seed, within-family differences; report family means and dispersion. Where sample size supports it, use paired bootstrap intervals over episode seeds while preserving family structure. Report raw counts when outcomes are sparse. A zero count in a small episode is not evidence of equivalence.

Decision Rule. The hypothesis is supported only if the full policy improves family-averaged MSMI over all three official baselines with zero invalid assessed episodes, and the gain is not explained by one favourable family, seed, or hidden coverage collapse.

Paired Comparison Workflow



STRESS TEST

Ablations, Failure Modes and Falsification

Ablation Ladder

VARIANT	QUESTION ISOLATED
Full core policy	Does the integrated core improve the primary metric?
Greedy allocation	Is joint allocation responsible for gain?
Baseline clarification	Does allocation-aware clarification improve decisions?
No clarification	Does clarification improve matching outcomes?
Direct MSMI model	Does the conditional funnel help?
Uncalibrated weights	Does held-out calibration improve decisions?
Exploration on/off	If considered, does bounded exploration repay its cost?
Waiting on/off	If considered, does strategic deferral add value?
Adaptation on/off	If considered, does recency adaptation help under drift?

Compact Experimental Matrix

VARIANT	REFERENCE	ONE CHANGE
Core full	-	Core policy; optional modules off
Greedy	Core full	Replace MWM with greedy
Base clarif.	Core full	Replace aVOI with baseline asks
No clarif.	Core full	Remove clarification asks
Direct	Core full	Replace funnel with direct model
Raw weights	Core full	Disable calibration
Explore on	Core full	Enable exploration
Wait on	Core full	Enable waiting
Adapt on	Core full, drift	Enable adaptation

Interpretation discipline. Core ablations change one core mechanism at a time. Optional modules are off by default; test an on/off pair only when that module is a candidate addition, using identical paired worlds.

Failure Register

RISK	FAILURE MECHANISM	MITIGATION / EVIDENCE
Sparse graph / nondisclosure	Hard constraints or declined fields eliminate supply.	Plausibly unlocking asks; never impute; report unserved members.
Miscalibration / sparse MSMI	Cardinal weights mislead matching and VOI.	Shrinkage, held-out reliability, direct-model comparator.
Selection bias / delays	Chosen pairs alone reveal outcomes; pending looks negative.	Diverse rollouts, maturity masks, no unsupported propensity correction.
Drift	Historical probabilities become stale.	Conservative released-feedback adaptation; off ablation.
Runtime spikes	Dense graphs and re-solves exceed limits.	Pruning, caching, caps, fallback, container profiling.
Waiting / coverage loss	Option value is overestimated.	No-wait default; horizon cap; service diagnostics.

What Would Falsify or Weaken the Hypothesis

The thesis is weakened if aVOI fails to beat baseline clarification, global matching does not beat greedy in paired rollouts, calibration does not improve decisions, gains vanish after coverage/runtime controls, or any assessed episode is invalid. It is falsified for this challenge if the full policy fails to outperform all three official baselines on the pre-specified family-averaged primary metric.

Pre-Specified Interpretation

RETAIN CORE
paired MSMI gain

REMOVE OPTIONAL
no consistent benefit

REJECT BUILD
invalid or over budget

SOURCES AND AUDIT TRAIL

References and Reproducibility

[1] Official challenge specification. The One Introduction Problem: Sequential Reciprocal Matching Under Uncertainty. Final Participant Specification 1.0.0, 5 Oct. 2026. Supplied archive. Official GitHub repository.

[2] Pizzato, L. A. S., Rej, T., Chung, T., Koprinska, I., and Kay, J. 'RECON: A Reciprocal Recommender for Online Dating.' ACM RecSys, 207-214, 2010. doi:10.1145/1864708.1864747.

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[5] Valancius, M., Lennon, M., and Oliva, J. 'Acquisition Conditioned Oracle for Nongreedy Active Feature Acquisition.' ICML / PMLR 235, 48957-48975, 2024. PMLR paper.

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[10] Niculescu-Mizil, A., and Caruana, R. 'Predicting Good Probabilities with Supervised Learning.' ICML, 625-632, 2005. doi:10.1145/1102351.1102430.

[11] Bifet, A., and Gavalda, R. 'Learning from Time-Changing Data with Adaptive Windowing.' SIAM SDM, 443-448, 2007. doi:10.1137/1.9781611972771.42.

Reproducibility and Compliance

Reproducibility. Freeze code, dependencies, model assets, declared training seeds and inference seed. Preserve pool separation, pre-action snapshots, clarification logs, maturity masks, scenario-level outputs and full episode JSON. Re-run tests, data verification and isolated container evaluation.

Compliance. All profiles and outcomes are synthetic. Inputs are limited to observable state, clarification results, released feedback, declared assets and policy-owned memory. No experimental scores, latency measurements or confidence intervals are asserted.

Version boundary. Mechanics were checked against the supplied official challenge archive. If organisers issue a later correction, the implementation and fixed submission version should record the governing specification explicitly.

Final Audit Checklist

AUDIT AREA	VERIFIED IN DESIGN	EVIDENCE TO RETAIN
State and interface	Batched asks; exact refresh; no future/private state.	Action-time snapshots, ask logs and memory resets.
Validity and allocation	Exact feasibility, non-overlap and bounded fallback.	Validity tests, solver status and fallback counts.
Evaluation	Mature labels; six families; three official baselines.	Paired seeds, scenario outputs and raw counts.